Pandas I: Introduction to Data Manipulation

Wook-Shin Han

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Learning Objectives

By the end of this lecture, you will be able to:

- Understand the concept of tabular data and its importance in data science
- Master the three fundamental pandas data structures: Series, DataFrame, and Index
- Create DataFrames using multiple methods
- Perform data selection and slicing operations
- Apply professional data manipulation techniques

What is Tabular Data?

Definition

Tabular data is information organized in a table format with rows and columns, representing one of the most common and flexible ways to store data in data science.

Structure

- **Rows**: Each row represents one observation or instance
- **Columns**: Each column represents a feature or characteristic
- **Flexibility**: Easy to understand, manipulate, and analyze

Real-World Example: Elections Dataset

- Each row: A presidential candidate in a specific year
- Each column: Candidate information
  - Year
  - Name
  - Party
  - Popular Vote
  - Result
  - Percentage

Why Pandas?

Industry Standard

Pandas is the accepted standard tool for tabular data manipulation in both industry and academia.

Key Capabilities

- Arrange data in tabular format
- Extract information with specific conditions
- Perform data operations for insights
- Apply NumPy functions efficiently
- Execute vectorized computations for speed

Three Fundamental Data Structures

1. Series

1-dimensional labeled array data

- Think of it as a single column
- Contains values of the same type
- Has an associated index (labels)

2. DataFrame

2-dimensional tabular data

- Collection of Series sharing the same Index
- Rows and columns structure
- Primary working data structure

3. Index

Sequence of row/column labels

- Can be integers, strings, or other types
- Does not need to be unique
- Provides data alignment and selection capabilities

Series Anatomy: index, name, values

Create a Series with labeled index and a Series name

```
s = pd.Series([4, 2, 4, 6],
    index=['cat', 'penguin', 'dog', 'butterfly'],
    name='legs')
s.index.name = 'animal'
```

Inspect the components

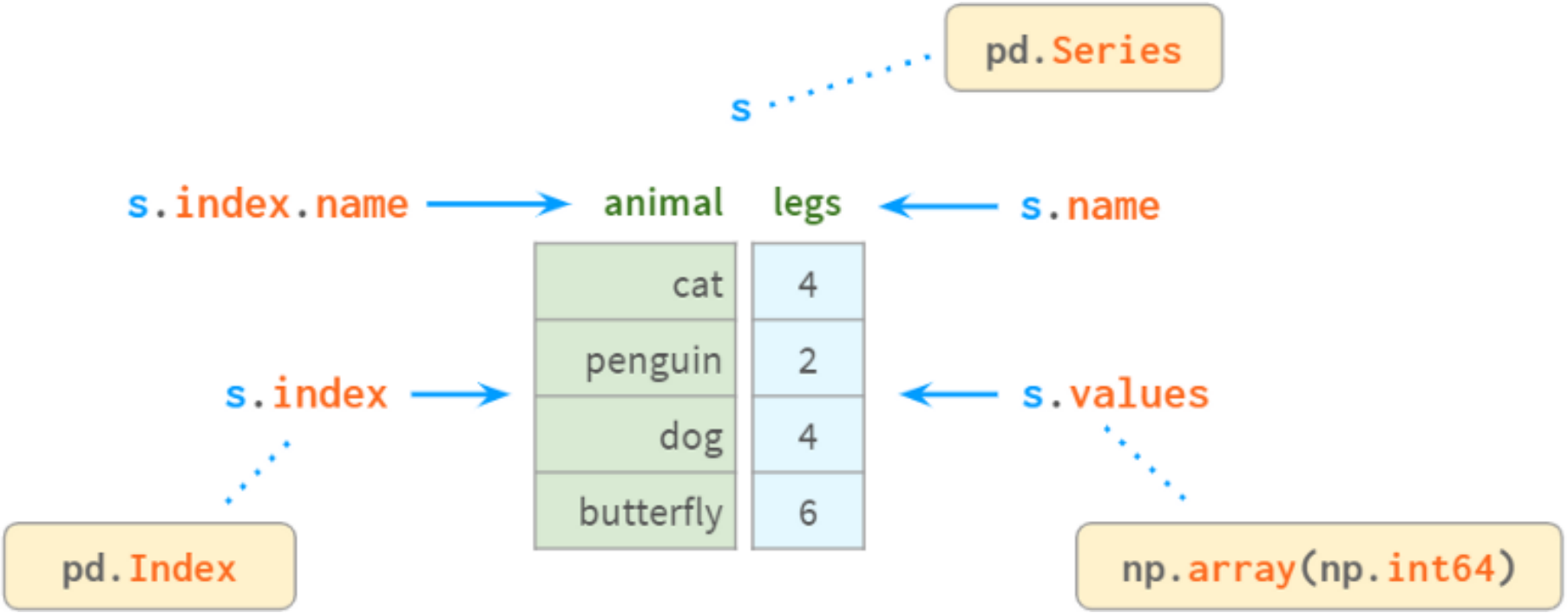
```
type(s)             # pandas.core.series.Series
s.index             # pd.Index([...], dtype='object', name='
    animal')
type(s.index)       # pandas.core.indexes.base.Index
s.index.name        # 'animal'  (label for the index axis)
s.name              # 'legs'    (label for the Series itself)
s.values            # np.ndarray([...], dtype=np.int64)
# Prefer s.to_numpy() for a consistent NumPy view across
    dtypes
```

- `s.index` holds the row labels (`pd.Index`); `s.index.name` is its title.
- `s.name` names the Series; useful after `reset_index()` or in joins.
- `s.values` returns a NumPy array; use `s.to_numpy()` when working with extension dtypes.

Working with Series — Core operations

- Series Index
- Series with Custom Index
- Series: Label vs. Position Indexing
- Boolean Filtering Idioms (`isin`, `between`)
- Take by Labels or Positions
- Index Edits: Assign, Sort, Reset, Delete
- Find Element by Value (Vector vs. Indexed Lookup)
- Missing Values: dtypes and `isna()`
- Missing Values II: Fill or Interpolate
- Missing Values III: Reductions with `NaN`
- Missing Values IV: Alignment in Arithmetic
- Sort, Rank, Top-k, Quantiles
- Cumulative, Difference, Lag, Return
- Where/Mask, Clip, Replace
- Map, Astype, To-NumPy
- Accessors: .str, .dt, .cat
- Windows: Rolling, Expanding, EWM
- Fast Scalars; Value Counts; Sampling

Series Index



- Index immutability: replace the whole index rather than editing in place, e.g. `s.index = pd.Index([...], name='animal')`.
- Check uniqueness: `s.index.is_unique`. If there are duplicates, `s.loc['label']` may return multiple rows.
- Types: values use `s.dtype`; labels use `s.index.dtype`. Mismatched dtypes often explain selection bugs.

Labeled Array Model vs Relational Model

- **Data abstraction.**
  - **Labeled array (pandas):** `pd.Series`/`pd.DataFrame` = arrays with *labels* (index, columns); order is meaningful and preserved.
  - **Relational (SQL):** relations (tables) as sets/bags of tuples with named attributes; logical order is irrelevant (order only via `ORDER BY`).
- **Identity & keys.**
  - Labeled array: `Index` provides labels (can be non-unique); behaves like an address space.
  - Relational: *keys* (PK/CK) identify tuples; constraints enforce uniqueness & referential integrity.
- **Missing data.**
  - Labeled array: `NaN`/`pd.NA`; many reductions use `skipna=True`.
  - Relational: `NULL` with three-valued logic; aggregates often ignore `NULL` by definition.
- **Core operations (mental mapping).**
  - `s1 + s2` (Series): *align by labels* then vectorized math ≈ `FULL OUTER JOIN` on keys then compute.
  - `df.merge` ≈ SQL `JOIN` (`inner`/`left`/`outer`/`cross`).
  - `df.groupby(...).agg(...)` ≈ SQL `GROUP BY` aggregates.
  - `pd.concat` (axis=0) ≈ `UNION ALL`; (axis=1) aligns on index like a join on row labels.

Setup

```
import pandas as pd
grades = pd.Series([85, 92, 78, 96],
                   index=['Alice', 'Bob', 'Charlie', 'Diana'
                       ],
                   name='grades')
```

- **Label-based:** `grades.loc[label / list / slice]`
  - Uses *index labels* (e.g., `'Alice'`).
  - Slices are *inclusive* of the stop label: `grades.loc['Alice':'Charlie']`.
  - Missing labels raise `KeyError`; use `reindex` if you want missing → `NaN`.
- **Position-based:** `grades.iloc[pos / list / slice]`
  - Uses *0..n-1* positions only.
  - Slices are *exclusive* of the stop: `grades.iloc[0:2]`.
  - Prefer `iloc` over `[]` when the index is integer-like to avoid ambiguity.
- **Boolean mask (conditional):** `grades[mask]` or `grades.loc[mask]`
  - `mask` is a same-length bool Series/array, e.g., `grades >= 90`.
  - Combine conditions with `&`, `|`, `~` and *parentheses*: `(grades >= 90) & (grades < 95)`.

Tip: For setting values, prefer `grades.loc[...]` on the left-hand side to avoid chained assignment.

Examples of Series Selection Methods

Setup

```
grades = pd.Series([85, 92, 78, 96],
                   index=['Alice', 'Bob', 'Charlie', 'Diana'
                       ],
                   name='grades')
```

Label-based: `grades.loc[label / list / slice]`

```
grades.loc['Alice']                 # 85
grades.loc[['Alice', 'Diana']]      # multiple labels
grades.loc['Alice':'Charlie']       # slice is INCLUSIVE of stop label
# grades.loc['Eve']                 # KeyError (use reindex to allow missing
    -> NaN)
```

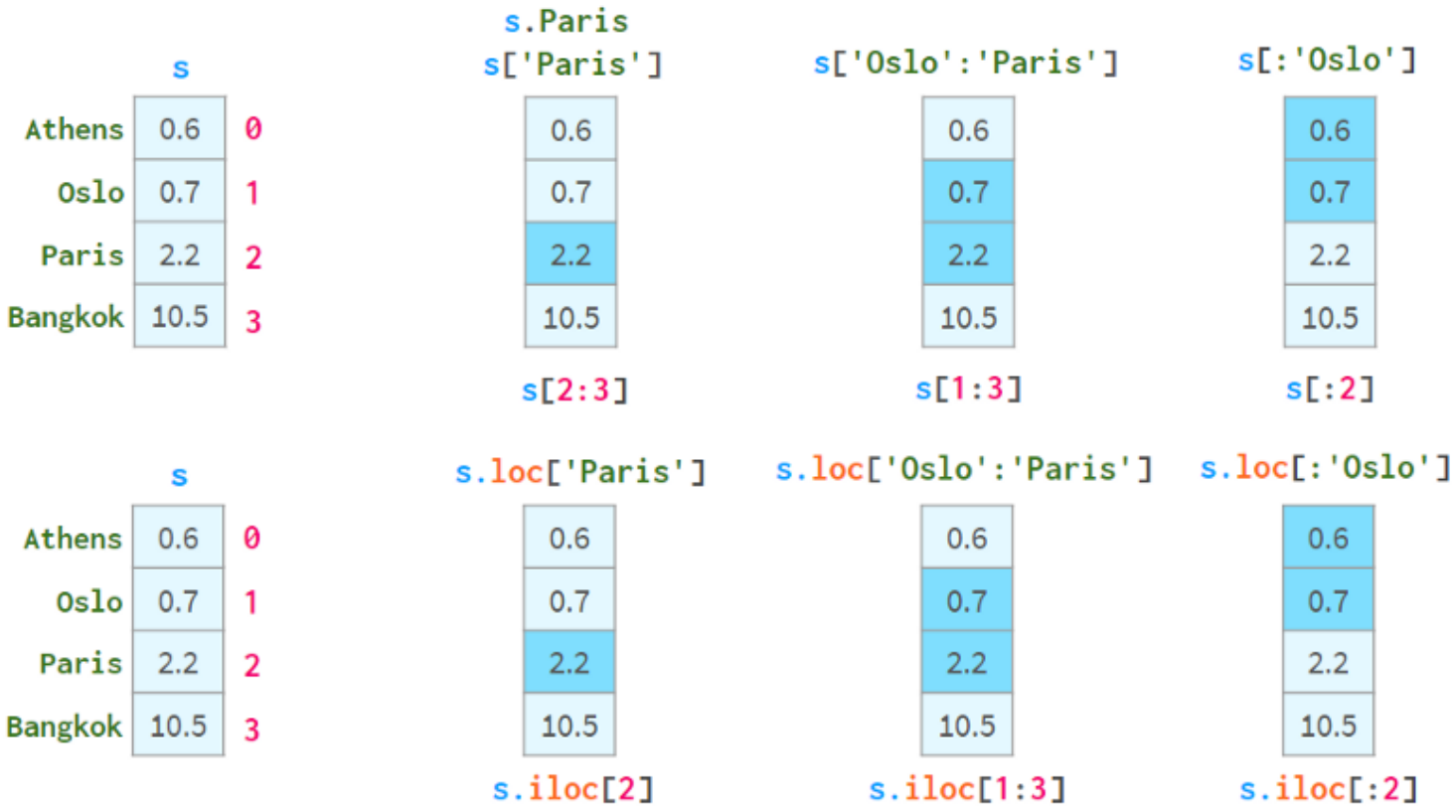
Position-based: `grades.iloc[pos / list / slice]`

```
grades.iloc[0]                      # 85
grades.iloc[[0, 3]]                 # positions list
grades.iloc[0:2]                    # slice is EXCLUSIVE of stop (Alice, Bob)
# Prefer iloc over [] when the index is integer-like (avoids ambiguity)
```

Boolean mask (conditional): `grades[mask]` **or** `grades.loc[mask]`

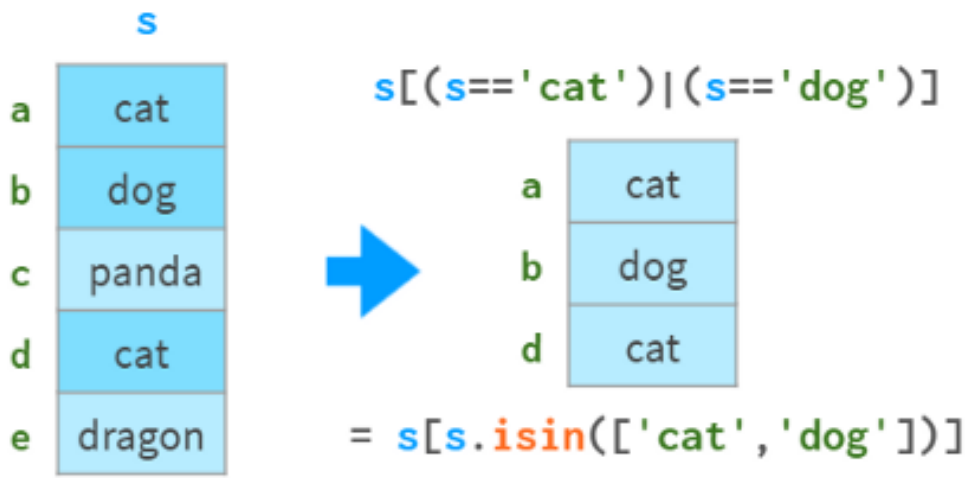
```
mask = grades >= 90                 # same-length boolean mask
grades[mask]                        # Bob, Diana
grades.loc[(grades >= 90) & (grades < 95)]   # combine with &, |, ~ and
    parentheses
```

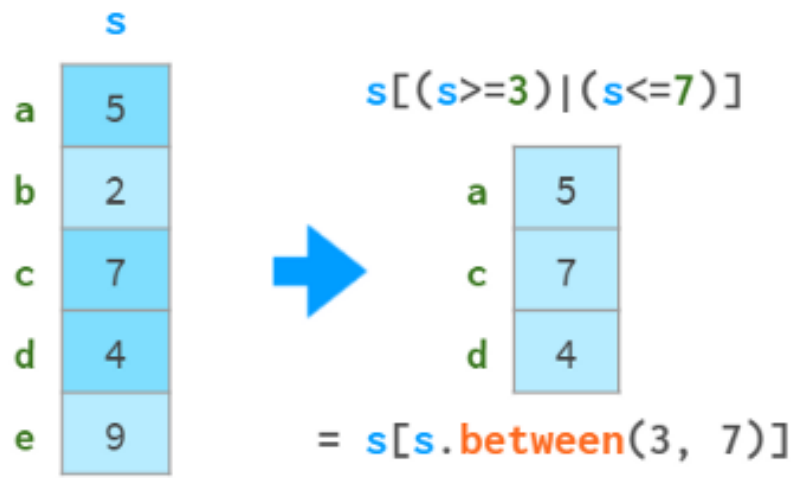
Series: Label vs. Position — why prefer `.loc`?



- **Explicitness:** `.loc` = labels, `.iloc` = positions. Plain `[]` is overloaded (`s['Paris']` label vs. `s[1:3]` positional slice).
- **Slice rule:** `loc` is *inclusive* of the stop; `iloc` is *exclusive*. Use `.loc`/`.iloc` to make intent clear.
- **No ambiguity with integers:** for integer-like indexes, `s[1]` means label 1. Use `.loc[1]` (label) or `.iloc[1]` (2nd element).

Boolean Mask on *Values* (`isin`, `between`)





- **Important:** `s == 'cat'` compares *values* element-wise, not labels. It yields a boolean mask the same length as `s`.
- Filter by *values*: `s[(s=='cat') | (s=='dog')]` ≡ `s[s.isin(['cat','dog'])]`.
- Filter by *labels* instead: `s.loc[s.index.isin(['a','d'])]` or `s.loc[s.index == 'a']`.
- Numeric ranges: use `&` (AND), not `|`; remember parentheses. `s[(s>=3) & (s<=7)]` ≡ `s.between(3, 7)` (inclusive by default).

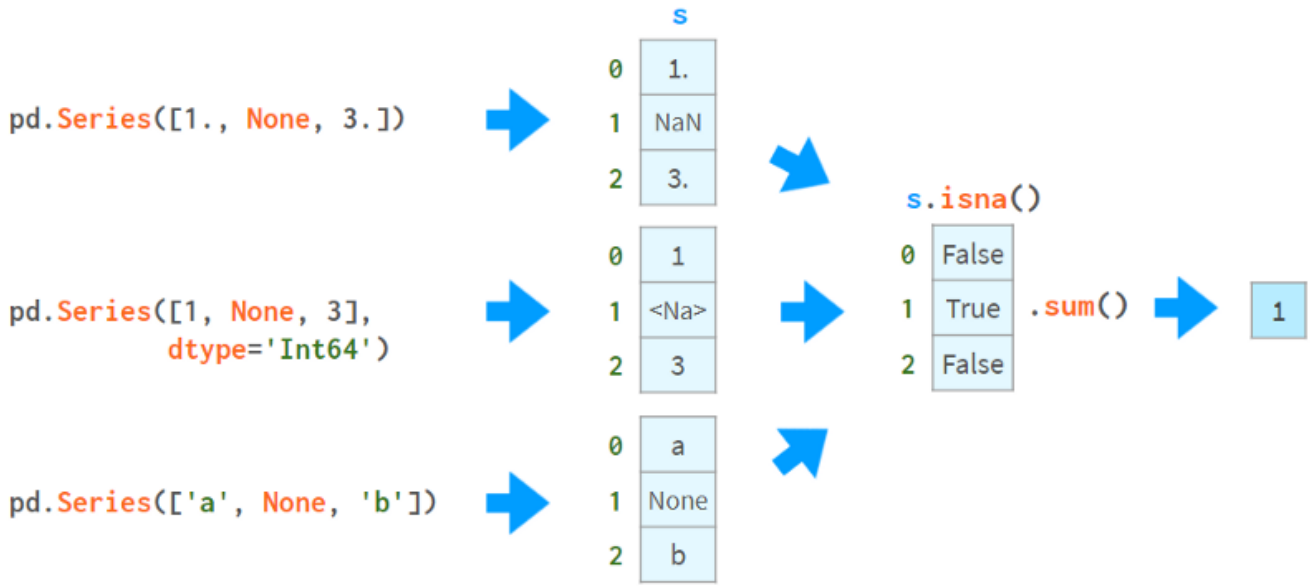
Find by Value — Vector scan vs Indexed lookup

- **Vector scan (values → labels/positions):**
  - `s.eq(v)`/`s==v` → mask, then `s.index[mask]` or `mask.to_numpy().nonzero()`.
  - Extremes: `s.idxmax()`/`s.idxmin()` return the *label* of max/min.
  - For ordered data, `s.searchsorted(v)` gives insertion position (on `s.values` or `s.sort_values()`).
- **Indexed lookup (keys → values):**
  - Normal case is `s.loc[key]`.
  - To map *value→label(s)* efficiently, build a reverse map once: `rev = pd.Series(s.index, index=s.values)` (handle duplicates separately).
- **Duplicates:** a value can occur multiple times; use group map to lists.
- **Pitfall:** value search is over *values*, not index labels; do not confuse with `.loc`.

Find by Value — Quick Examples

```
import pandas as pd
s = pd.Series([85, 92, 78, 92, 96], index=['Alice','Bob','Charlie','Eve','
    Diana'])

# 1) Vector scan: who scored 92?
who = s.index[s.eq(92)].tolist()          # ['Bob','Eve']

# 2) Label of max / min
top, low = s.idxmax(), s.idxmin()         # ('Diana', 'Charlie')

# 3) Reverse map (one-to-one only!)
rev = pd.Series(s.index, index=s.values)
rev.loc[96]                               # 'Diana'
# For duplicates -> list map:
rev_multi = s.reset_index().groupby(0)['index'].apply(list)
rev_multi.loc[92]                         # ['Bob','Eve']

# 4) Position(s) of a value
import numpy as np
pos = np.flatnonzero(s.to_numpy() == 92)  # array([1, 3])
```

Missing Values — dtypes and `isna()`

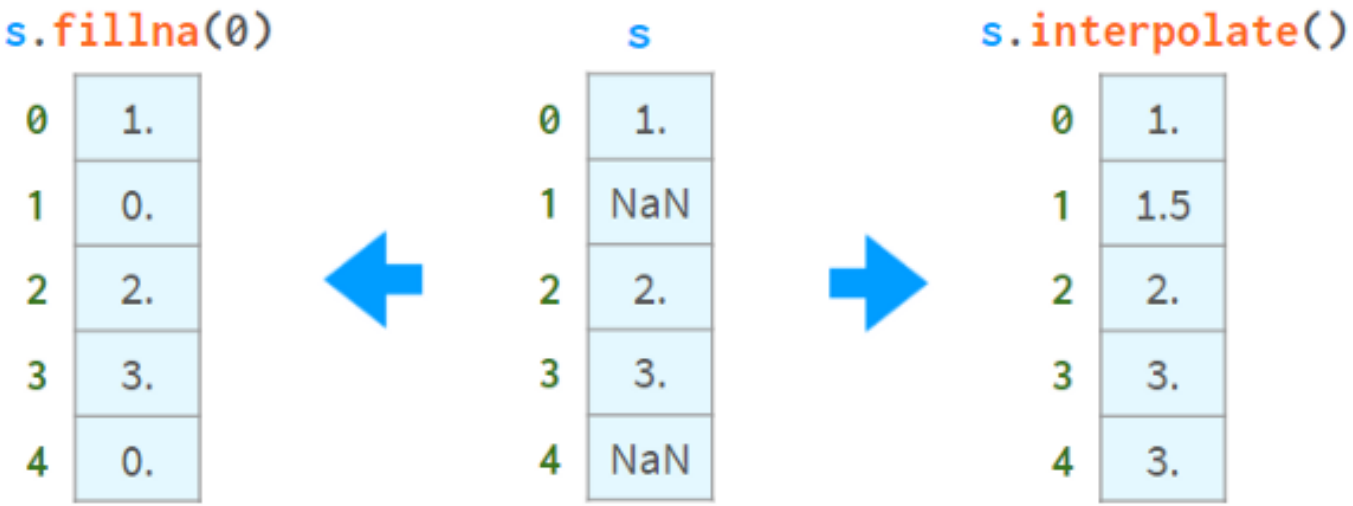


- Missing sentinels: `np.nan` (NumPy float NaN), `pd.NA` (pandas scalar for NA-aware dtypes).
- NA-aware dtypes: `Int64`, `Float64`, `boolean`, `string`; preserve type with missing via `pd.NA`. (i.e., three-valued logic)
- NumPy dtypes → upcasting: mixing ints with `np.nan` yields `float64`.
- Detect missing: `s.isna()` → `pd.Series[bool]`, aliases: `pd.isna`, `s.notna()`.
- Counting: `s.isna().sum()` (# missing), `s.count()` (# non-missing).
- Construction tips: use `dtype='Int64'` or `'boolean'` to avoid float upcast when you need integers/booleans with NA.

isna() and dtypes — Quick Examples

```
import numpy as np, pandas as pd

# Upcasting with NumPy NaN
s = pd.Series([1, 2, np.nan, 4])
s.dtype
# dtype('float64')

s.isna()
# 0    False
# 1    False
# 2     True
# 3    False
# dtype: bool

# NA-aware integer dtype keeps integers+NA
t = pd.Series([1, pd.NA, 3], dtype="Int64")
t, t.dtype
# 0       1
# 1    <NA>
# 2       3
# dtype: Int64

# DataFrame-wide missing map and counts
df = pd.DataFrame({"a":[1, pd.NA, 3], "b":[np.nan, 2.0, 3.0]})
df.isna()
df.isna().sum()       # per column
df.notna().all()      # no missing in a column?
```

Missing Values II — Fill or Interpolate

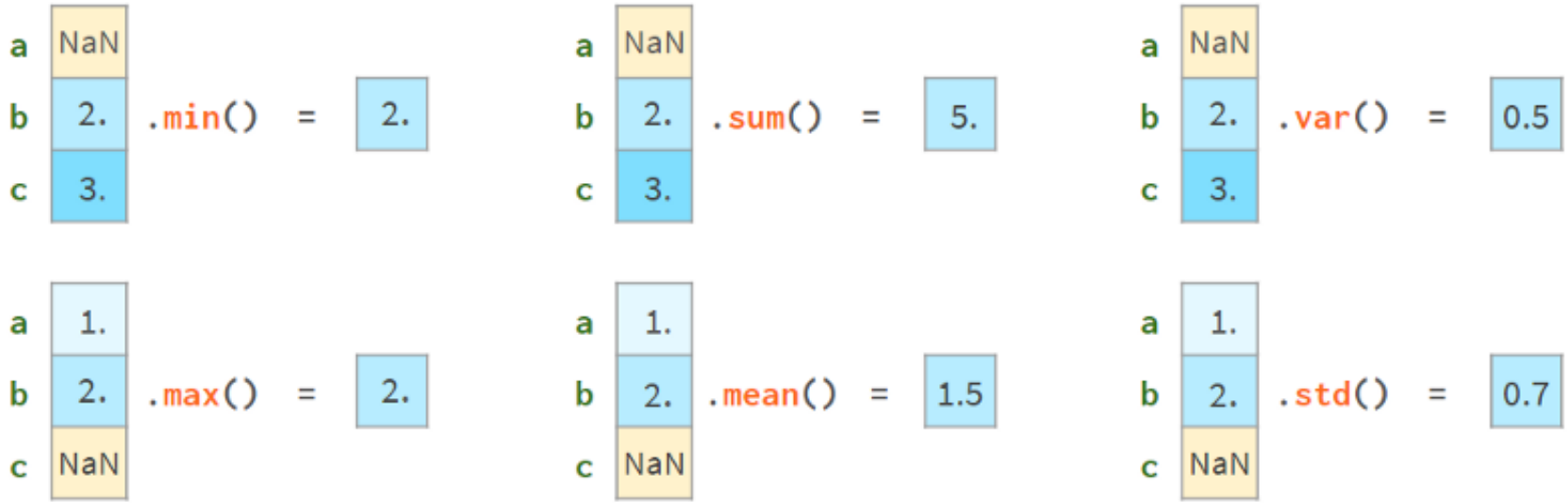


- **Direct fill:** `s.fillna(value)`; `df.fillna({col: val})`. Methods: `method='ffill'|'bfill'`, with `limit=` to cap consecutive fills.
- **Propagate last/next:** `s.ffill()` fill missing values with the last seen valid value (forward carry), `s.bfill()` (backward carry).
- **Interpolation:** `s.interpolate(method='linear'|'index'|'time', limit_direction=...)`. Requires monotonic index for `'index'`/`'time'` (DatetimeIndex for `'time'`).
- **When to use what:** constants/sentinels ⇒ `fillna`; carry last valid ⇒ `ffill/bfill`; trend-aware gaps ⇒ `interpolate`.
- **Caution:** Interpolation estimates values; document your choice of `method`, `limit`, and whether time spacing is uniform.

fillna, ffill/bfill, interpolate — Quick Examples

```
import pandas as pd, numpy as np

s = pd.Series([1.0, np.nan, np.nan, 4.0, np.nan])

s.fillna(0)
# 1.0, 0.0, 0.0, 4.0, 0.0

s.ffill()
# 1.0, 1.0, 1.0, 4.0, 4.0

s.bfill(limit=1)
# 1.0, nan, 4.0, 4.0, nan   # only first gap filled due to limit=1

# Time-aware interpolation
idx = pd.date_range("2024-01-01", periods=5, freq="D")
t = pd.Series([1.0, np.nan, 4.0, np.nan, 9.0], index=idx)
t.interpolate(method="time")
# linearly interpolates by actual time spacing

# Per-column fills with a dict
df = pd.DataFrame({"a":[1, np.nan, 3], "b":[np.nan, 2.5, np.nan]})
df.fillna({"a": 0, "b": df["b"].mean()})
```

Missing Values III — Reductions with NaN

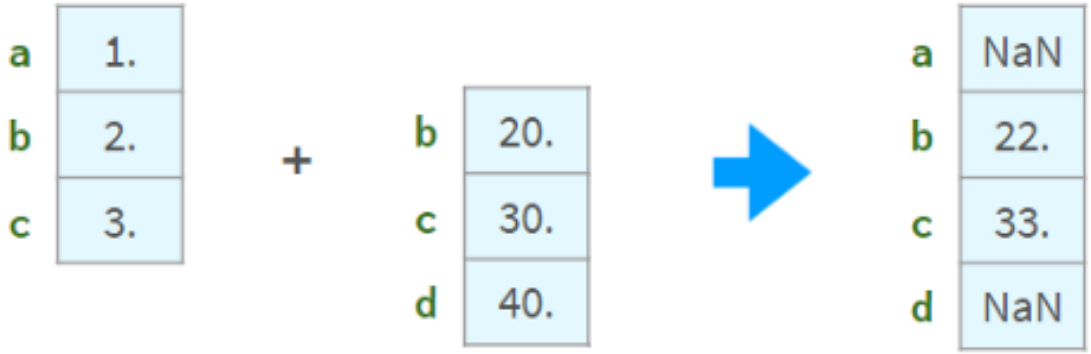


- NA-skipping by default: `s.sum()/mean()/min()/max()` use `skipna=True` (default). Control via `skipna=False` to require complete data.
- All-missing behavior: `s.mean()` → `NaN`; `s.sum()` → `0` (for numeric). `s.count()` counts non-missing elements.
- Robust counts: `s.isna().sum()` (# missing)
- Uniques with NA: `s.nunique(dropna=True)` ignores missing; set `dropna=False` to include.
- Boolean reductions: `s.any()` / `s.all()` ignore missing with `skipna=True`; use `skipna=False` to propagate `NaN`.

Reductions with NaN — Quick Examples

```
import numpy as np, pandas as pd

s = pd.Series([1.0, np.nan, 3.0, np.nan])

s.sum(), s.mean(), s.count()
# (4.0, 2.0, 2)      # sum skips NaN; mean over non-missing; count=2

s.mean(skipna=False)
# nan                # requires complete data

pd.Series([np.nan, np.nan]).sum(), pd.Series([np.nan, np.nan]).mean()
# (0.0, nan)

df = pd.DataFrame({"a":[1, np.nan, 3], "b":[2, 4, np.nan]})
df.mean(axis=0), df.mean(axis=1)
# a->2.0, b->3.0     # by column
# row means with NaNs skipped

# Boolean reductions
b = pd.Series([True, pd.NA, False], dtype="boolean")
b.any(), b.all()
# (True, False)      # NA ignored under skipna=True
```

Missing Values IV — Alignment in Arithmetic



- **Index alignment:** binary ops align on labels; union of keys is used. Non-overlapping labels → `NaN` in result.
- **Method forms with fill:** `s1.add(s2, fill_value=0)`, `df1.sub(df2, fill_value=0)`, `mul/div/pow` analogous.
- **DataFrame alignment:** aligns on both index and columns. Use `df1.add(df2, fill_value=0)` to avoid `NaN` where only one side has data.
- **Broadcasting with alignment:** `df.add(s, axis='index'|'columns')` matches along the given axis labels.
- **Preparation:** `reindex` to a common index/columns (optionally with `fill_value`) before arithmetic for deterministic outcomes.

Example of Alignment in Arithmetic

Code

```python
import pandas as pd

s1 = pd.Series({"a": 1.0, "b": 2.0})
s2 = pd.Series({"b": 10.0, "c": 100.0})

s1 + s2
s1.add(s2, fill_value=0)
```

Output

```
# s1 + s2
# a      NaN
# b     12.0
# c      NaN
# dtype: float64

# s1.add(s2, fill_value=0)
# a       1.0
# b      12.0
# c     100.0
# dtype: float64
```

Sort, Rank, Top-k, Quantiles

- `s.sort_values(ascending=...)` → `pd.Series`: reorder by *values*; stable within ties; preserves index.
- `s.sort_index(ascending=...)` → `pd.Series`: reorder by *labels* (index); stable; preserves index.
- `s.rank(method='dense', ascending=...)` → `pd.Series[float]`: assign ranks; `dense` gives 1,2,3,... even with ties.
- `s.nlargest(k)` → `pd.Series` / `s.nsmallest(k)` → `pd.Series`: fastest top-k/bottom-k by values (returned Series is sorted).
- `s.quantile(q)` → `float` | `pd.Series`: scalar if q is a single float in [0,1]; Series (indexed by q) if list-like.

Setup for examples

```
import pandas as pd
s = pd.Series([5, 2, 7, 4, 9], index=list('abcde'))
# a:5, b:2, c:7, d:4, e:9
```

Examples of Sort, Rank, Top-k, Quantiles

```
s.sort_values(ascending=False)
# e     9
# c     7
# a     5
# ...

s.sort_index()
# a     5
# b     2
# c     7
# ...

s.rank(method='dense', ascending=False)
# a     3.0
# b     5.0
# ...
# e     1.0

s.nlargest(2)
# e     9
# c     7

s.nsmallest(2)

s.quantile([0.25, 0.50, 0.75])
# 0.25    4.0
# 0.50    5.0
# 0.75    7.0
```

Cumulative, Difference, Lag, Return

- `s.cumsum()` $\rightarrow$ `pd.Series`: cumulative sum.
- `s.cummax()` $\rightarrow$ `pd.Series`: running maximum.
- `s.cummin()` $\rightarrow$ `pd.Series`: running minimum.
- `s.cumprod()` $\rightarrow$ `pd.Series`: cumulative product.
- `s.shift(periods=k, fill_value=None)` $\rightarrow$ `pd.Series`: **move values** by k rows while keeping the same index (lag if $k > 0$, lead if $k < 0$); first/last $|k|$ positions become `NaN` unless `fill_value` is given.
- `s.diff(periods=k)` $\rightarrow$ `pd.Series[float]`: **change vs. k-step lag**; exactly $s - s.\text{shift}(k)$. First k values are `NaN`; negative k computes a forward difference.
- `s.pct_change(periods=k)` $\rightarrow$ `pd.Series[float]`: **relative change** vs. k-step lag; exactly $\frac{s}{s.\text{shift}(k)} - 1$. First k values are `NaN`; beware division by zero $\rightarrow$ `inf`/`NaN`.

diff, shift, pct_change — Quick Examples

```
s = pd.Series([100, 110, 121], index=['t0','t1','t2'])

s.shift(1)                    # move values down by 1 (lag)
# t0      NaN
# t1    100.0
# t2    110.0

s.diff(1)                     # == s - s.shift(1)
# t0     NaN
# t1    10.0      (110 - 100)
# t2    11.0      (121 - 110)

s.pct_change(1)               # == s / s.shift(1) - 1
# t0       NaN
# t1      0.10     (110/100 - 1)
# t2      0.10     (121/110 - 1)

s.diff(2)                     # difference vs. two steps back
# t0     NaN
# t1     NaN
# t2    21.0       (121 - 100)

s.shift(-1)                   # lead: pull next value up
# t0    110.0
# t1    121.0
# t2      NaN
```

Tips:

- Use shift(k, fill_value=...) to avoid leading/trailing NaN

Where/Mask, Clip, Replace — What they do

- `s.where(cond, other)` → `pd.Series`: keep values where `cond`, else `other`.
- `s.mask(cond, other)` → `pd.Series`: inverse of `where`.
- `s.clip(lower, upper)` → `pd.Series`: clamp to bounds.
- `s.replace(to_replace, value)` → `pd.Series`: value remapping.

Where/Mask, Clip, Replace — Two-Column Examples

Code (s setup + ops)

```python
# Set up the Series s
s = pd.Series([5, -2, 7, -1, 9],
              index=list("abcde"))

# Where: keep >= 0, else 0
s_where = s.where(s >= 0, 0)

# Mask: set < 0 to 0 (inverse of where)
s_mask = s.mask(s < 0, 0)

# Clip: bound values to [0, 6]
s_clip = s.clip(lower=0, upper=6)

# Replace specific values
s_repl = s.replace({-2: 0, -1: 0})
```

Output

```
s_where
a    5
b    0
c    7
d    0
e    9

s_mask
a    5
b    0
c    7
d    0
e    9

s_clip
a    5
b    0
c    6
d    0
e    6

s_repl
a    5
b    0
c    7
d    0
e    9
```

Map, Astype, To-NumPy — What they do

- `s.map(func | dict)` → `pd.Series`: elementwise transform or dictionary lookup (index preserved). Unmatched keys become `NaN`; use `fillna(...)` if needed.
- `s.astype(dtype)` → `pd.Series`: cast values to a new dtype (e.g., `'float64'`, `'Int64'`, `'string'`). Fails if coercion is impossible; use `errors='ignore'` to skip.
- `s.to_numpy(copy=False)` → `np.ndarray`: NumPy view/array of the values. Prefer over `s.values` (works consistently with extension dtypes).

Map, Astype, To-NumPy — Two-Column Examples

Code

```
import pandas as pd

code2name = {'US':'United States', 'KR'
    'Korea'}
country = pd.Series(['US','KR','XX'])
country_name = country.map(code2name).
    fillna('Unknown')

s = pd.Series([5, 2, 7], index=list('abc
    ')).astype('float64')
a = s.to_numpy(copy=False)
```

Output

```
country_name
# 0    United States
# 1            Korea
# 2          Unknown
# dtype: object

s
# a    5.0
# b    2.0
# c    7.0
# dtype: float64

a
# array([5., 2., 7.])
```

Accessors: .str, .dt, .cat — What they do

- `s.str.method(...)` → `pd.Series[string|object]`: vectorized string ops (normalize, split, extract). Requires string/nullable string dtype for best behavior.
- `s.dt.method(...)` → `pd.Series`: datetime parts (`year`, `month`), conversions (`to_period`), floor/ceil/round. Requires datetime dtype.
- `s.astype('category').cat.method(...)` → `pd.Series | Categorical`: add/remove categories, order, encode. Categories are fixed—add with `add_categories` before assignment.

Accessors: .str, .dt, .cat — Two-Column Examples

Code

```python
import pandas as pd

email = pd.Series(['A@x.com','B@Y.com',
    None], dtype='string')
domain = email.str.strip().str.lower().
    str.split('@').str[-1]

dt = pd.to_datetime(pd.Series(['
    2025-09-01','2025-09-12']))
years   = dt.dt.year
periods = dt.dt.to_period('M')

cat = pd.Series(['a','b','a'], dtype='
    category')
cat  = cat.cat.add_categories(['c'])
    # add an unused 'c'
cat2 = cat.cat.remove_unused_categories
    ()
```

Output

```
domain
# 0     x.com
# 1     y.com
# 2      <NA>
# dtype: string

years
# 0     2025
# 1     2025
# dtype: int64

periods
# 0     2025-09
# 1     2025-09
# dtype: period[M]

cat2
# 0     a
# 1     b
# 2     a
# dtype: category
# Categories (2, object): ['a', 'b']
```

Windows: Rolling, Expanding, EWM — What they do (and why those numbers?)

Given $s = [5, 2, 7, 4, 9]$ with index a, b, c, d, e

- **Rolling mean** `s.rolling(w, min_periods).mean()` → `pd.Series[float]` Fixed-size window of length w. At each index t, average the last w points (fewer if #obs < `min_periods`). $\mathrm{rollMean}_w[t] = \frac{1}{n_t} \sum_{i=t-n_t+1}^{t} s[i]$ where $n_t = \min(w, \#\text{seen})$.
  *Examples (*w=3, `min_periods=1`*):* $b : \frac{5+2}{2} = 3.5,\ c : \frac{5+2+7}{3} = 4.6667,\ \cdots$
- **Expanding sum** `s.expanding().sum()` → `pd.Series` Growing window from the start (cumulative). $\mathrm{expandSum}[t] = \sum_{i \le t} s[i]$. *Examples:*
  $c : 5 + 2 + 7 = 14,\ e : 5 + 2 + 7 + 4 + 9 = 27.$
- **EWMA (adjust=False)** `s.ewm(alpha, adjust=False).mean()` → `pd.Series[float]` Exponentially weighted moving average (more weight on recent). $y_t = \alpha\, x_t + (1 - \alpha)\, y_{t-1}$ with $y_0 = x_0$. *For* $\alpha = 0.5$*:* $a : 5.0000$ (seed), $b : 0.5 \cdot 2 + 0.5 \cdot 5 = 3.5000,\ c : 0.5 \cdot 7 + 0.5 \cdot 3.5 = 5.2500,$ $d : 0.5 \cdot 4 + 0.5 \cdot 5.25 = 4.6250,\ e : 0.5 \cdot 9 + 0.5 \cdot 4.625 = 6.8125.$

Notes: `center=True` centers the rolling window; `min_periods` controls early `NaN`s; α can be given via `span/com/halflife`.

Example: Windows: Rolling, Expanding, EWM

Code

```
import pandas as pd
s = pd.Series([5, 2, 7, 4, 9], index=
    list('abcde'))

roll3 = s.rolling(3, min_periods=1).mean
    ().round(4)
expd  = s.expanding().sum()
expo  = s.ewm(alpha=0.5, adjust=False).
    mean().round(4)
```

Output (with why)

```
roll3
# a  5.0000   # (5)/1
# b  3.5000   # (5+2)/2
# c  4.6667   # (5+2+7)/3
# d  4.3333   # (2+7+4)/3
# e  6.6667   # (7+4+9)/3

expd
# a   5       # 5
# b   7       # 5+2
# c  14       # 5+2+7
# d  18       # 5+2+7+4
# e  27       # 5+2+7+4+9

expo
# a  5.0000   # seed: y0 = x0 = 5
# b  3.5000   # 0.5*2 + 0.5*5
# c  5.2500   # 0.5*7 + 0.5*3.5
# d  4.6250   # 0.5*4 + 0.5*5.25
# e  6.8125   # 0.5*9 + 0.5*4.625
```

Fast Scalars; Value Counts; Sampling — What they do

- `s.at[label]` → `scalar`: fast scalar by label (index lookup); raises `KeyError` if missing.
- `s.iat[pos]` → `scalar`: fast scalar by position (0-based); raises `IndexError` if out of bounds.
- `s.value_counts(dropna=..., normalize=...)` → `pd.Series`: frequency table (or proportions with `normalize=True`); `dropna=False` to count missing.
- `s.sample(n | frac, random_state)` → `pd.Series`: random rows (without replacement by default). Use `replace=True` for bootstrap-like sampling.

Fast Scalars; Value Counts; Sampling — Two-Column Examples

Code

```
import pandas as pd

s    = pd.Series([5, 2, 7, 4, 9], index=
     list('abcde'))
vals = pd.Series(['x','x','y', None, 'x'
     ])

c = s.at['c']; p = s.iat[2]
counts = vals.value_counts(dropna=False)
props  = vals.value_counts(normalize=
     True, dropna=False).round(2)
one    = s.sample(n=1, random_state=0)
```

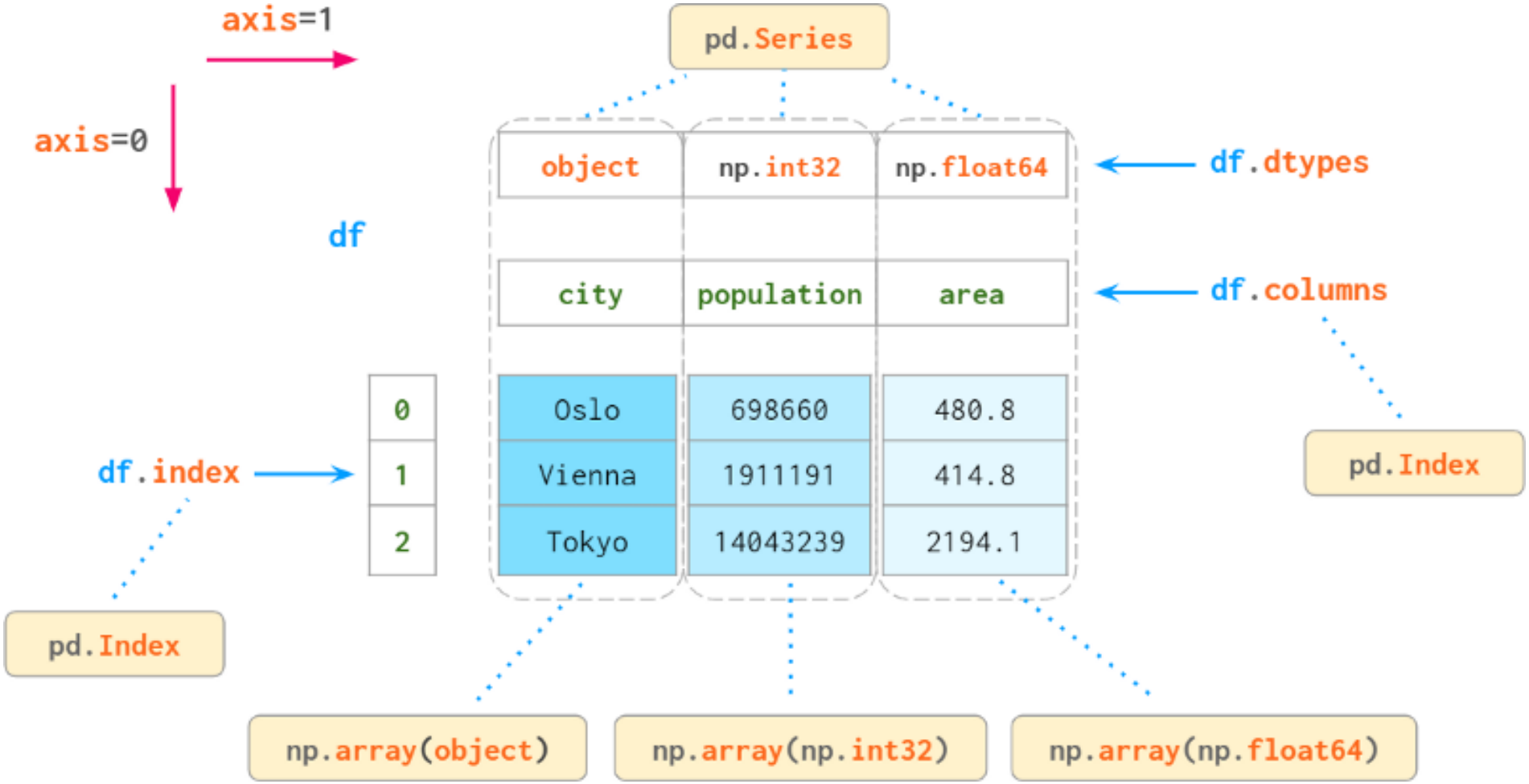
Output

```
(c, p)
# (7, 7)

counts
# x       3
# y       1
# NaN     1
# dtype: int64

props
# x    0.60
# y    0.20
# NaN  0.20
# dtype: float64

one
# e    9
# dtype: int64
```

Working with DataFrames — Core operations

- Anatomy & Diagnostics: `df.index`, `df.columns`, `df.dtypes`, `df.info()`, `df.select_dtypes(...)`
- Column Picks: `df['col']` (Series) vs. `df[['col']]` (DataFrame); `df.assign(new=...)`
- Label vs. Position Indexing: `df.loc[row_labels, col_labels]` (inclusive slices), `df.iloc[row_pos, col_pos]` (exclusive)
- Fast Scalars: `df.at[row, col]`, `df.iat[i, j]`
- Boolean Filtering: masks with `&`, `|`, `~`; `.isin()`, `.between()`, `.query(...)`
- Across-Columns Tests: `df.isna().any(axis=1)`, `df.gt(0).all(axis=1)`
- Safe Assignment: use `.loc[mask, 'col']` on LHS; `.copy()` after subsetting; align with `reindex`
- Missing Values: `df.isna()`, `fillna({col:val})`, `ffill/bfill` (axis=0/1), `interpolate`
- Alignment & Broadcasting: binary ops align on both axes; `df.add/ sub/ mul(..., fill_value=0)`; `add(series, axis='index'|'columns')`
- Sorting & Top-k: `df.sort_values(by=[...])`, `df.sort_index()`, `df.nlargest(k, 'col')/nsmallest`
- Reshape (Index/Columns): `set_index/reset_index`, `pivot/pivot_table`, `melt` (wide↔long)

DataFrame Anatomy: axes, columns, dtypes, index



- **Axes:** `axis=0` (rows), `axis=1` (columns).
- **Labels:** `df.columns` (column labels), `df.index` (row labels, a `pd.Index`).
- **Types:** each column is a `pd.Series` with its own `dtype`; see `df.dtypes`.
- Conversions to NumPy: `df['col'].to_numpy()` or `df.to_numpy()`.

DataFrame Anatomy — Python code example

```
# Recreate the example table with explicit dtypes
df = pd.DataFrame({
    "city":       pd.Series(["Oslo", "Vienna", "Tokyo"], dtype="object"),
    "population": pd.Series([698660, 1911191, 14043239], dtype=np.int32),
    "area":       pd.Series([480.8, 414.8, 2194.1],     dtype=np.float64),
})

# Inspect axes (axis=0 rows, axis=1 columns) and labels
print("index (axis=0)   :", df.index)      # RangeIndex(0, 3)
print("columns (axis=1) :", df.columns)    # Index([...], dtype='object')

# Each column is a Series with its own dtype
print("\nColumn dtypes:\n", df.dtypes)

# Conversions to NumPy (1D for a Series, 2D for a DataFrame)
print("\ncity -> numpy:", df["city"].to_numpy())                # dtype=
    object
print("pop+area -> numpy:\n", df[["population","area"]].to_numpy()  #
    numeric 2D

# Simple axis demo on numeric columns
num = df[["population","area"]]
print("\nColumn-wise sum (axis=0):\n", num.sum(axis=0))
print("\nRow-wise sum (axis=1):\n", num.sum(axis=1))
```

DataFrame Creation Methods

From CSV File

```
# Reading elections dataset
elections = pd.read_csv("elections.csv")
print(elections.head(2))
#    Year          Candidate                Party  Popular vote Result
# 0  1824     Andrew Jackson  Democratic-Republican       151271   loss
# 1  1824  John Quincy Adams  Democratic-Republican       113142    win
```

From Dictionary

```
student_data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Math': [85, 92, 78],
    'Science': [88, 95, 82]
}
df = pd.DataFrame(student_data)
print(df)
#      Name  Math  Science
# 0   Alice    85       88
# 1     Bob    92       95
# 2 Charlie    78       82
```

Read CSV: header and data

cities.csv:

csv = comma-separated values

```
name, population, area
Oslo, 698660, 480.8
Vienna, 1911191, 414.8
Tokyo, 14043239, 2194.1
```

header

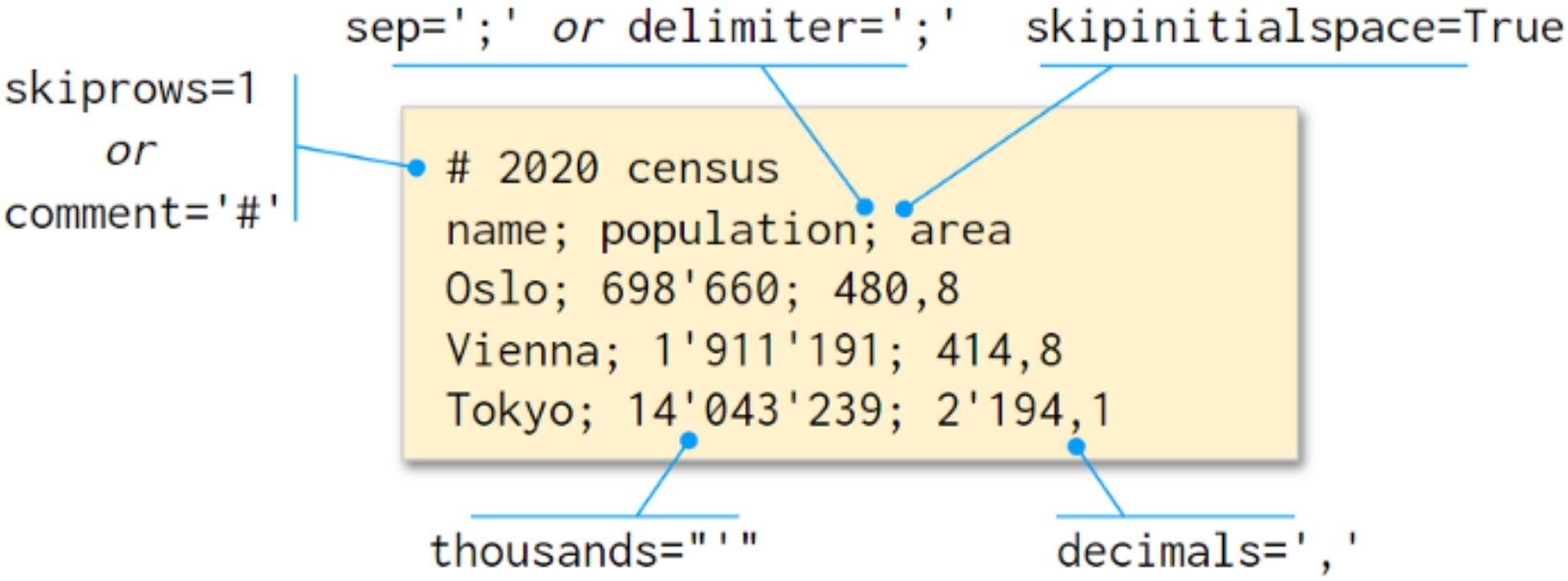
data

```
df = pd.read_csv('cities.csv')
```

	name	population	area
0	Oslo	698660	480.8
1	Vienna	1911191	414.8
2	Tokyo	14043239	2194.1

- Basic load: `df = pd.read_csv('cities.csv')` (first row becomes header by default).
- Comma-separated values; pandas infers dtypes per column.
- Use `header=None` if the file has no header row.

CSV Parsing Options (locales, comments, spacing)



- Field separator: `sep=';'` (or `delimiter=';'`).
- European formats: `thousands="'"` (apostrophe), `decimal=','`.
- Ignore spaces after separators: `skipinitialspace=True`.
- Skip lines: `skiprows=1` or treat # as comments with `comment='#'`.

read_csv / to_csv: essential arguments

```
• pd.read_csv(
      filename,                 # build a dataframe out of this csv-file
      sep=',',                  # or delimiter=',' - field separator
      skipinitialspace=False,   # whether to allow spaces after sep
      quotechar='"',            # used for quoting strings that contain sep
      skiprows=n,               # skip the first n lines (default is 0)
      usecols=columns,          # only read those columns
      nrows=n,                  # only read n rows (useful for reading in chunks)
      index_col=columns,        # use column(s) as an index
      ...)                      # ~50 more less frequently used arguments

• df.to_csv(filename, ...)      # write data from the dataframe in a csv format
```

- `pd.read_csv(filename, sep=',', skipinitialspace=False, quotechar='"', skiprows=n, usecols=cols, nrows=n, index_col=cols, ...)`.
- Write out: `df.to_csv(filename, index=True/False, sep=',', na_rep='')`.
- Tip: use `usecols` to read a subset; `nrows` to sample; `index_col` to set the index on load.

Set the Index on Read vs. After Read

```
df = pd.read_csv('cities.csv')
```

	name	population	area
0	Oslo	698660	480.8
1	Vienna	1911191	414.8
2	Tokyo	14043239	2194.1

```
df.set_index('name')
```

```
df = pd.read_csv('cities.csv',
        index_col='name')
```

name	population	area
Oslo	698660	480.8
Vienna	1911191	414.8
Tokyo	14043239	2194.1

- Two equivalent patterns: `df = pd.read_csv('cities.csv'); df = df.set_index('name')` vs. `df = pd.read_csv('cities.csv', index_col='name')`.
- Setting `index_col` at read-time avoids an extra copy.

More DataFrame Creation Examples

From Lists (2D)

```
# Creating DataFrame from nested lists
data = [['Alice', 85, 88], ['Bob', 92, 95], ['Charlie', 78, 82]]
df = pd.DataFrame(data, columns=['Name', 'Math', 'Science'])
```

From Series

```
# Combining Series into DataFrame
names = pd.Series(['Alice', 'Bob', 'Charlie'])
math_scores = pd.Series([85, 92, 78])
df = pd.DataFrame({'Name': names, 'Math': math_scores})
```

DataFrame Attributes with Examples

```
# Create sample DataFrame
df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Math': [85, 92, 78],
    'Science': [88, 95, 82]
})

print("Index:", df.index)
# Index: RangeIndex(start=0, stop=3, step=1)

print("Columns:", df.columns)
# Columns: Index(['Name', 'Math', 'Science'], dtype='object')

print("Shape:", df.shape)
# Shape: (3, 3)

print("Data types:")
print(df.dtypes)
# Name        object
# Math         int64
# Science      int64
# dtype: object
```

Data Selection: Head and Tail

First n Rows

```
df.head(n)  # Default n=5
```

Last n Rows

```
df.tail(n)  # Default n=5
```

Use Cases

- Initial data exploration
- Verification after data operations
- Quick quality checks
- Presentation of sample data

Label-Based Selection: .loc Examples

```
# Sample data
df = elections.head()

# Single value selection
candidate = df.loc[0, 'Candidate']
print(candidate)  # Andrew Jackson

# Multiple rows, single column
candidates = df.loc[0:2, 'Candidate']
print(candidates)
# 0       Andrew Jackson
# 1    John Quincy Adams
# 2       Andrew Jackson
# Name: Candidate, dtype: object

# Multiple rows and columns (inclusive slicing)
subset = df.loc[0:2, 'Year':'Party']
print(subset)
#    Year         Candidate                Party
# 0  1824    Andrew Jackson  Democratic-Republican
# 1  1824  John Quincy Adams  Democratic-Republican
# 2  1828    Andrew Jackson            Democratic
```

More .loc Examples

```
# Select all rows, specific columns
winners = df.loc[:, ['Candidate', 'Result']]
print(winners)
#          Candidate Result
# 0    Andrew Jackson   loss
# 1  John Quincy Adams    win
# 2    Andrew Jackson    win

# Conditional selection with .loc
winners_only = df.loc[df['Result'] == 'win', ['Candidate', 'Year']]
print(winners_only)
#          Candidate  Year
# 1  John Quincy Adams  1824
# 2    Andrew Jackson  1828

# Boolean indexing
democratic = df.loc[df['Party'].str.contains('Democratic')]
print(democratic.shape)  # (3, 6)
```

Position-Based Selection: .iloc Examples

```
# Position-based selection (remember: exclusive slicing)
df = elections.head()

# Single value by position
first_candidate = df.iloc[0, 1]  # Row 0, Column 1
print(first_candidate)  # Andrew Jackson

# First 3 rows, first 4 columns (exclusive of endpoint)
subset = df.iloc[0:3, 0:4]
print(subset)
#    Year         Candidate                Party  Popular vote
# 0  1824    Andrew Jackson  Democratic-Republican      151271
# 1  1824  John Quincy Adams  Democratic-Republican      113142
# 2  1828    Andrew Jackson            Democratic      642806

# Last 2 rows, all columns
last_rows = df.iloc[-2:, :]
print(last_rows.shape)  # (2, 6)

# Every other row, specific columns
every_other = df.iloc[::2, [1, 4]]  # Columns 1 and 4
```

Context-Dependent Selection: []

Row Slicing

```
df[0:4]  # First four rows
```

Column Selection (List)

```
df[['Year', 'Candidate', 'Party']]  # Multiple columns
```

Single Column Selection

```
df['Candidate']  # Returns Series
```

Professional Usage

Most commonly used selector due to conciseness, especially for column selection operations.

Practical Example: Data Analysis Workflow

```
# Step 1: Load and inspect data
elections = pd.read_csv("elections.csv")
print(f"Dataset shape: {elections.shape}")
print(f"Columns: {list(elections.columns)}")

# Step 2: Basic exploration
print("First few rows:")
print(elections.head(3))

# Step 3: Specific analysis
print("Recent winners (2020-2024):")
recent_winners = elections[
    (elections['Year'] >= 2020) &
    (elections['Result'] == 'win')
][['Year', 'Candidate', 'Party']]
print(recent_winners)

# Step 4: Summary statistics
print(f"Total elections covered: {elections['Year'].nunique()}")
print(f"Unique candidates: {elections['Candidate'].nunique()}")
```

Indexing Methods Summary

Method	Use Case	Slicing	Input Type
`.loc`	Label-based selection	Inclusive	Labels
`.iloc`	Position-based selection	Exclusive	Integers
`[]`	Context-dependent	Varies	Mixed

Best Practices

Best Practices

- Use `.loc` for label-based operations (safer)
- Use `.iloc` for position-based logic
- Use `[]` for simple column selection
- Always consider data ordering when using `.iloc`

Professional Development Tips

Documentation Skills

- Pandas documentation is extensive
- Stack Overflow provides practical solutions
- Google search skills are essential
- Reading and implementing documentation is a core data science skill

Learning Approach

- Start with fundamental concepts
- Practice with real datasets
- Build familiarity through repetition
- Reference materials will be provided for assessments

Summary

Key Takeaways

1. **Tabular data** is fundamental to data science work
2. **Three core structures**: Series, DataFrame, Index
3. **Multiple creation methods** for different data sources
4. **Three selection approaches**: .loc, .iloc, []
5. **Professional practice** requires documentation skills

Next Steps

- Practice with real datasets
- Explore pandas documentation
- Prepare for Pandas II advanced operations
- Build comfort with data selection patterns

Questions and Discussion

Prepare for hands-on practice with real-world datasets in the next session